

Texas A&M University
CSCE 703 — Cybersecurity

Homework 2B

OWASP Juice Shop: Secure Design, Login Form, and Exploitation

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Part 1 — Secure Feature Design

I explored the live Juice Shop at preview.owasp-juice.shop and used what I found to design a safer registration/login flow. Three challenges stood out. **Login Admin** (Injection) shows how a crafted email can break a string-built SQL login query. Mitigation: parameterized queries or an ORM, plus email validation before any database call—so input stays data, not query code. **DOM XSS** shows how HTML/JavaScript can run in another user’s browser. Mitigation: encode output, prefer `textContent` over `innerHTML`, and add a Content-Security-Policy. **Forged Signed JWT** / weak authentication shows token handling can be abused for account takeover. Mitigation: verify JWT signatures with a strong secret, short expiry, server-side authorization, and rate-limited login. For registration, passwords must be stored with `bcrypt`, never plaintext.

Learnings (~100 words)

What surprised me most is how “normal” the Juice Shop login screen looks while still being easy to abuse if the backend concatenates SQL. Looking up Login Admin and DOM XSS on the Score Board made the risks concrete instead of abstract. I also realized authentication bugs are not only about passwords—token handling matters just as much. My takeaway for a secure registration/login design is simple: treat every field as hostile, keep queries parameterized, never render raw HTML from users, and hash passwords with `bcrypt` before they touch storage. Those few habits would have blocked most of what I saw in the challenges.

Password hashing example

```
const hash = await bcrypt.hash(password, 12);  
const ok = await bcrypt.compare(submittedPassword, user.passwordHash);
```

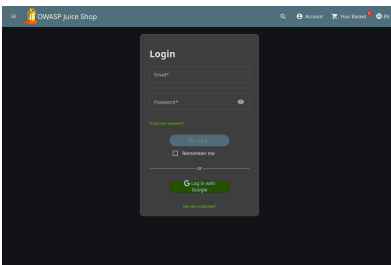


Figure 1. Juice Shop login.

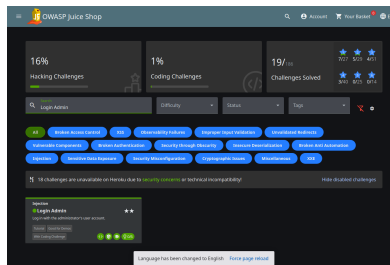


Figure 2. Login Admin.

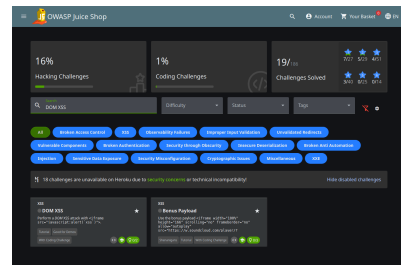


Figure 3. DOM XSS (JWT challenges listed nearby).

Part 2 — Front-End Login Form

I built a simple login page in HTML, CSS, and JavaScript with a small Express server behind it. The form has email and password fields. On the client, app.js blocks empty submissions, checks that the email contains “@”, and requires the password to be at least eight characters. On the server, those checks run again more strictly, passwords are verified with bcrypt, and login also requires a CSRF token from /api/csrf. Status messages use textContent so reflected text cannot turn into HTML. I kept demo credentials in the README only, not on the page.

Learnings (~100 words)

Building the form made the Juice Shop lessons feel practical. Client-side checks are nice for users—they catch empty fields quickly—but they are easy to skip, so I treated the server as the real gate. Adding CSRF forced me to think about forged requests, not only bad passwords. Hashing with bcrypt was straightforward once I stopped thinking of “encryption” and started thinking of one-way hashing. Overall I learned that a “basic” login page still needs several layers: validation, safe rendering, password hashing, and request authenticity. The public repo and README document how to run everything locally.

Public GitHub repository: <https://github.com/deeps45/juice-guard>

Run: `npm install && npm start` → `http://127.0.0.1:3847`

Demo login (README only): `demo@juice.shop` / `JuiceShop1!`

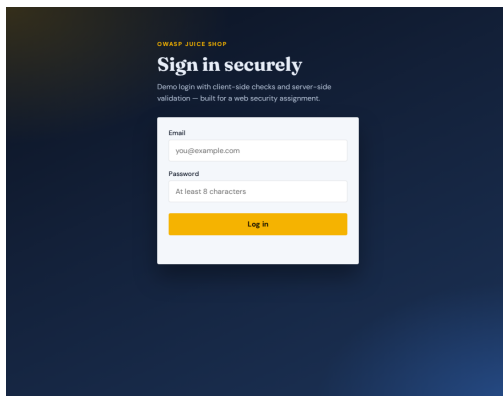
A screenshot of the login form titled "Sign in securely" with the subtitle "Demo login with client-side checks and server-side validation — built for a web security assignment." The form has two input fields: "Email" with the value "you@example.com" and "Password" with the value "At least 8 characters". Below the password field is a yellow "Log in" button.

Figure 4. My hardened login form.

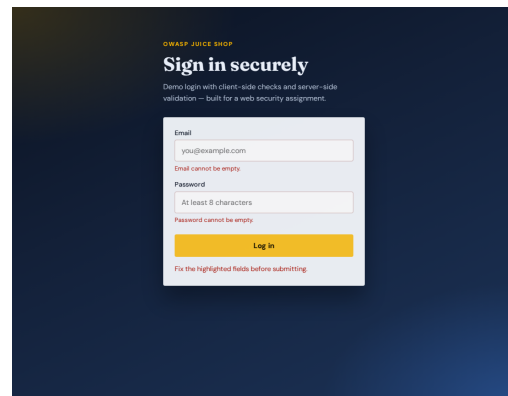
A screenshot of the login form showing validation errors. The "Email" field has a red border and the text "Email cannot be empty" below it. The "Password" field has a red border and the text "At least 8 characters" and "Password cannot be empty" below it. A yellow "Log in" button is at the bottom. A red message at the bottom says "Fix the highlighted fields before submitting."

Figure 5. Empty fields blocked.

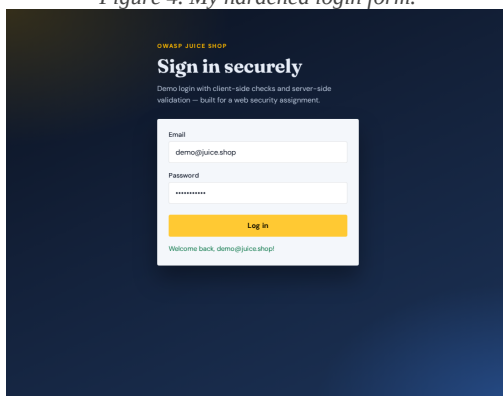
A screenshot of the login form showing a successful login. The "Email" field has the value "demo@juice.shop" and the "Password" field has the value "*****". Below the password field is a yellow "Log in" button. Below the button, a message says "Welcome back, demo@juice.shop!"

Figure 6. Successful login.

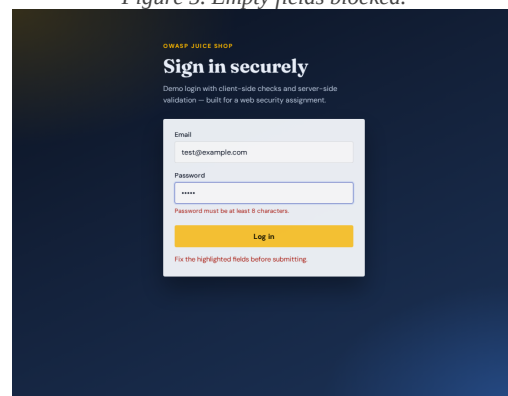
A screenshot of the login form showing a password validation error. The "Email" field has the value "test@example.com" and the "Password" field has the value "****". Below the password field is a red message that says "Password must be at least 8 characters." A yellow "Log in" button is at the bottom. A red message at the bottom says "Fix the highlighted fields before submitting."

Figure 7. Short password rejected.

Part 3 — Breaking My Own Form

I tried to break the same form that is in my GitHub repo. First I entered `xss@test.com` and the password `<script>alert(1)</script>`. The client actually allowed the submit (it only checks “@” and length), and the browser sent `POST /api/login`—but the server rejected unsafe characters and no script ran. Next I bypassed the UI with `fetch()` and got the same server error. A login `POST` without an X-CSRF-Token returned 403. I also tried `admin@test.com' OR '1'='1`; that was blocked as an invalid email. So classic XSS/SQLi did not fully land, but I did expose a real weakness: the client is too trusting on its own.

Learnings (~100 words)

The biggest lesson was that “the attack failed” is not the whole story. My client validation let a script-shaped password through and still called the API. That is exactly how people get hurt when they assume the browser is enough. Calling the endpoint with `fetch()` made the same point louder. CSRF testing showed why a random token belongs on state-changing requests. I fixed the gaps by keeping strict server checks, requiring CSRF tokens, using `textContent`, and setting a CSP. If I build another auth form, I will design the server path first and treat client checks as UX only.

Steps and results

XSS attempt: open / → email `xss@test.com` → password `<script>alert(1)</script>` → Log in. *Result:* client submitted the request; server returned an unsafe-character error; no alert.

Client bypass: `fetch('/api/login')` with the same password and a CSRF token. *Result:* HTTP 400 from the server.

CSRF check: `POST` valid credentials with no X-CSRF-Token. *Result:* HTTP 403.

SQLi probe: email `admin@test.com' OR '1'='1`. *Result:* rejected; no auth bypass.

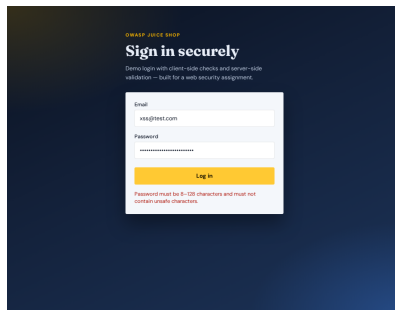


Figure 8. XSS blocked by server.

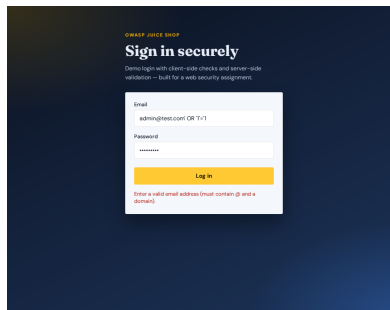


Figure 9. SQLi probe rejected.

Weaknesses found in my login form

Tests run against the real `/api/login` endpoint (not a mocked page).

1) Missing CSRF token → request rejected

Forgot login `POST` with valid domain credentials but no X-CSRF-Token:

```
HTTP/1.1 403
{
  "type": "https://www.ietf.org/rfc/rfc2616.txt#section-10.4.7",
  "error": "Missing or invalid CSRF token."
}
```

Result: CSRF weakness is mitigated → forged request blocked (403).

2) Client-side validation bypass → server still blocks XSS

Called `/api/login` directly with `fetch()`, skipping `validateEmail()`, using password `<script>alert(1)</script>`:

```
HTTP/1.1 400
{
  "type": "https://www.ietf.org/rfc/rfc2616.txt#section-10.4.7",
  "error": "Password must be 8-128 characters and must not contain unsafe characters."
}
```

Result: Client checks are bypassable, but server validation exposed the payload.

Figure 10. CSRF 403 + fetch bypass.

One fix I applied on top of server validation was requiring CSRF tokens and a Content-Security-Policy, so forged requests and inline scripts are harder to abuse even if a UI bug shows up later.